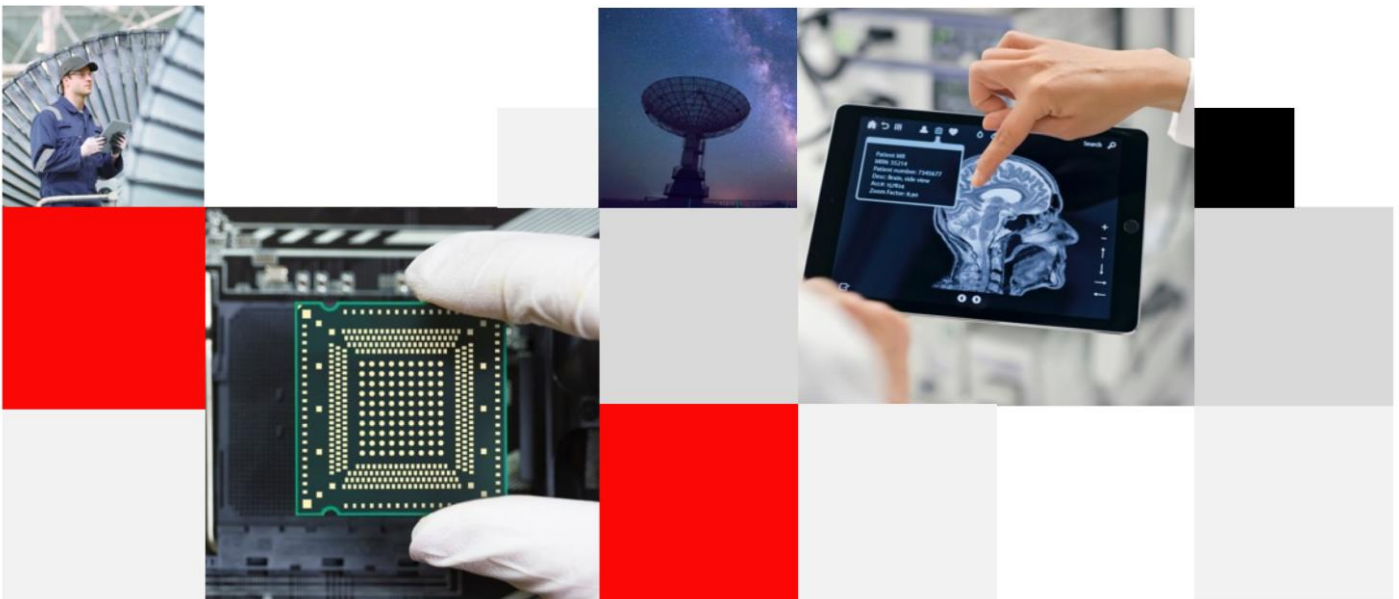


Wrote by	Approved by	Approved by
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Design Review Report

Rev.Design_Review_Report_V1.1

Date: 2025-09-29



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About the Document

History

Revision	Date	Author	Description
1.0	2025-09-19	Vinson/ Haikal	Initial Version
1.1	2025-09-29	Vinson/ Haikal	Second Review

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1 Scope

The aim of this document is to describe suggestions and corrections that Quectel advises to improve application that integrates Quectel EG800K module.

2 Design Review

Design review is based on the following received documentation:

- RoviGPS_EG800K_Design Review Report_V2.pdf

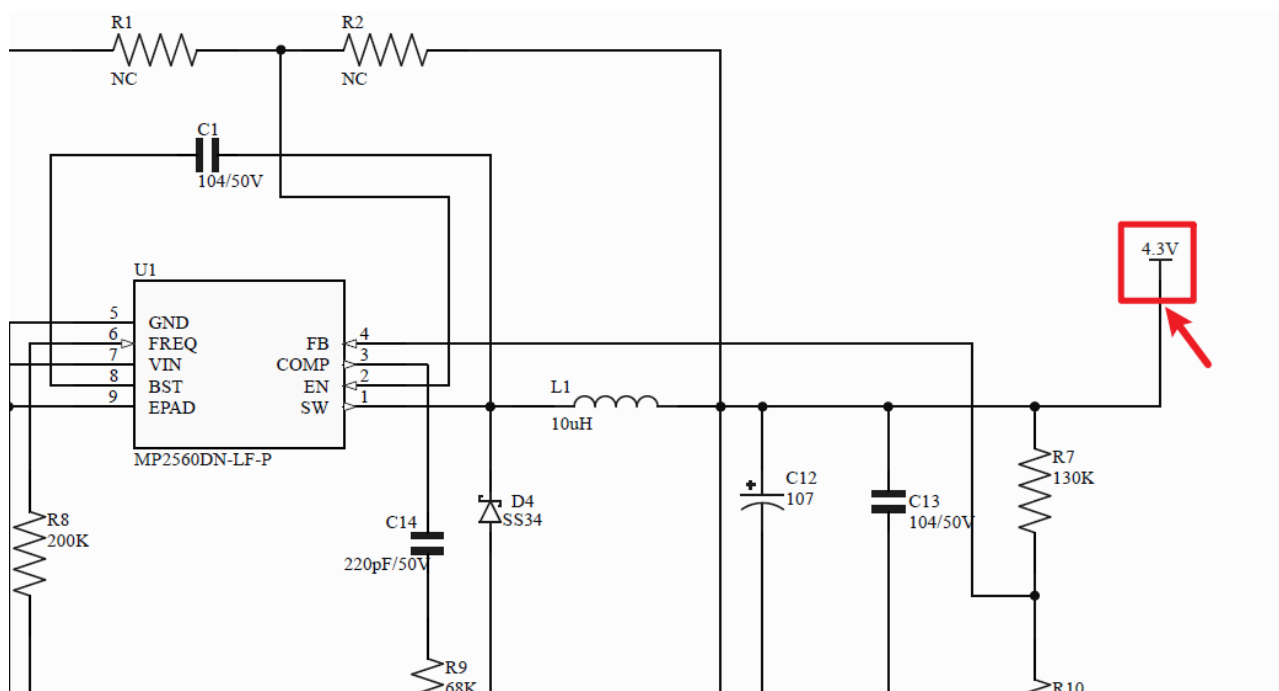
3 Schematic Review

3.1. POWER

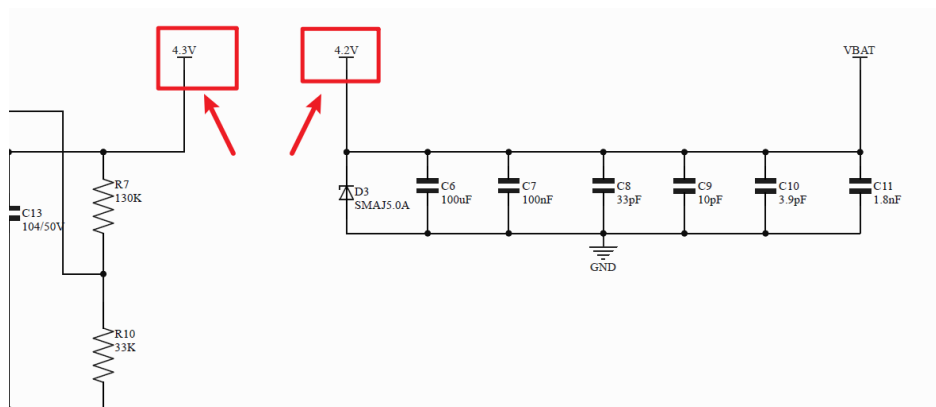
1. Please note that if V_{fb} is 0.8V, the calculated V_{out} would be approximately 3.9V.
Kindly ensure that the regulator calculation is correct on your side.

Thus the output voltage is:

$$V_{OUT} = V_{FB} \frac{(R1 + R2)}{R2}$$

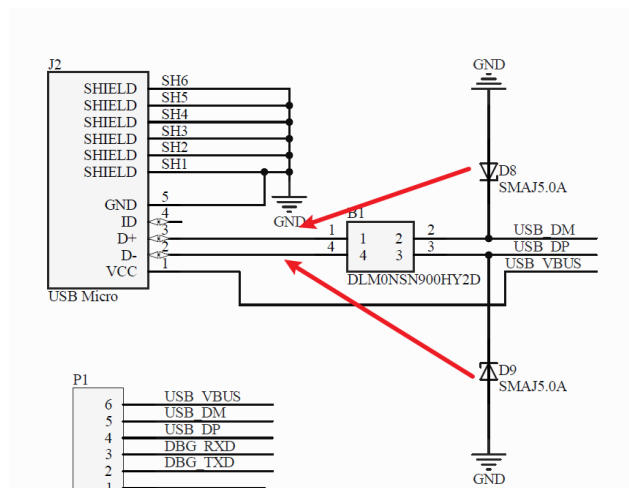


2. Please note that the voltage from the regulator to the supply source is inconsistent. Kindly correct this discrepancy.



3.2. USB

- To protect the USB interface against ESD, it is recommended to add a TVS diode on the VBUS, DM and DP lines. Additionally, the junction capacitance of the ESD protection devices on the USB data traces should be less than 2 pF. The TVS should place close to connector.



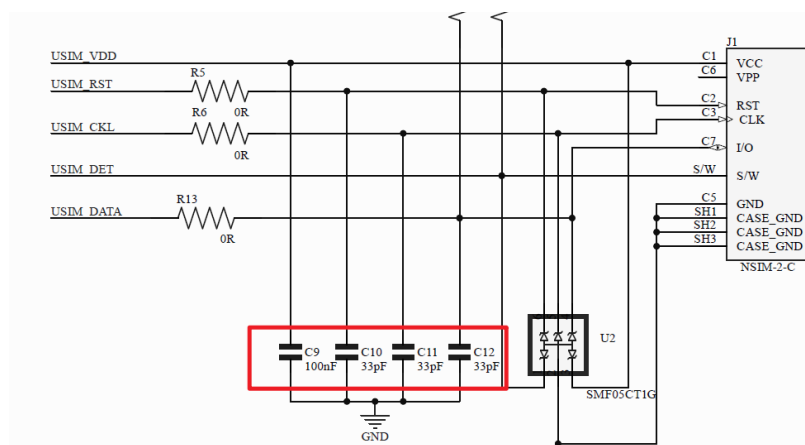
3.3. SIM CARD

- It is recommended to use a 15kΩ for the pull-up resistor on USIM_DATA instead of 51kΩ.
- To offer good ESD protection, it is recommended to add an ESD protection component whose parasitic capacitance should not be more than 15 pF. Please note that SMF05CT1G is exceed the limit.

SMF05CT1G ELECTRICAL CHARACTERISTICS ($T_J = 25^\circ\text{C}$ unless otherwise specified)

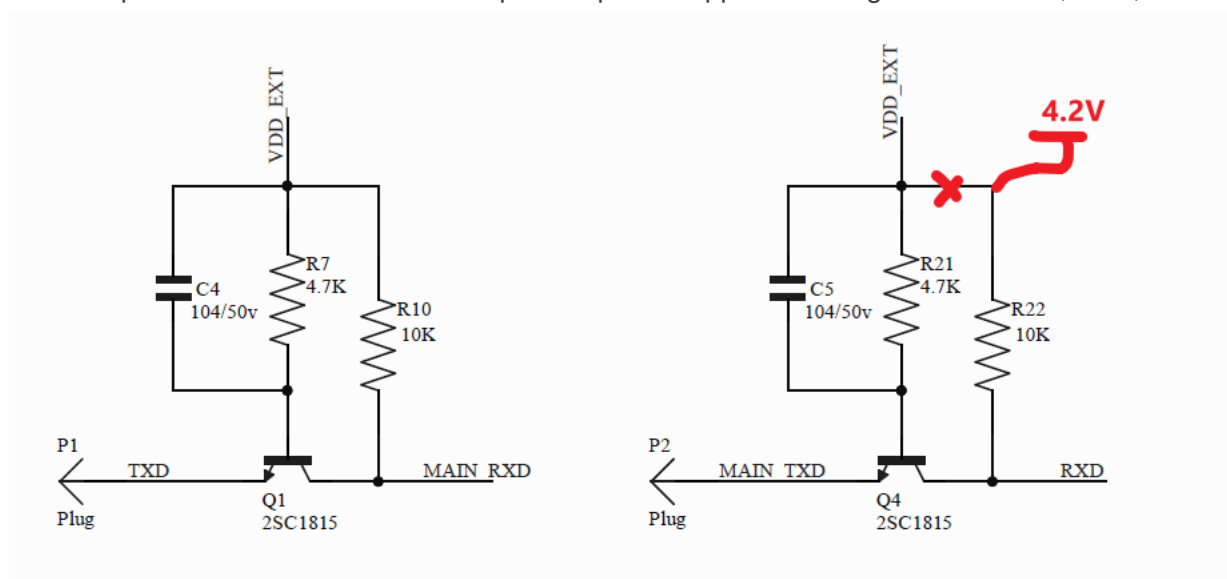
Parameter	Symbol	Conditions	Min	Typ	Max	Unit
Reverse Working Voltage	V_{RWM}	(Note 2)			5.0	V
Breakdown Voltage	V_{BR}	$I_T = 1\text{ mA}$, (Note 3)	6.2		7.2	V
Reverse Leakage Current	I_R	$V_{RWM} = 5\text{ V}$		0.07	5.0	μA
Clamping Voltage	V_C	$I_{PP} = 5\text{ A}$ (8 x 20 μs Waveform)			9.8	V
Clamping Voltage	V_C	$I_{PP} = 8\text{ A}$ (8 x 20 μs Waveform)			12.5	V
Maximum Peak Pulse Current	I_{PP}	8 x 20 μs Waveform			8.0	A
Capacitance	C_J	$V_R = 0\text{ V}$, $f = 1\text{ MHz}$ (Line to GND)		80	130	pF

- Please note that the previously used values for the filtering and bypass capacitors were correct. Kindly revise them back to the original values.



3.4. UART

1. Please note that previously the design is correct, however, the RXD signal should not pulled up to 1.8V. It should instead be pulled up to a supported voltage level of 3.3V, 4.2V, or 5V

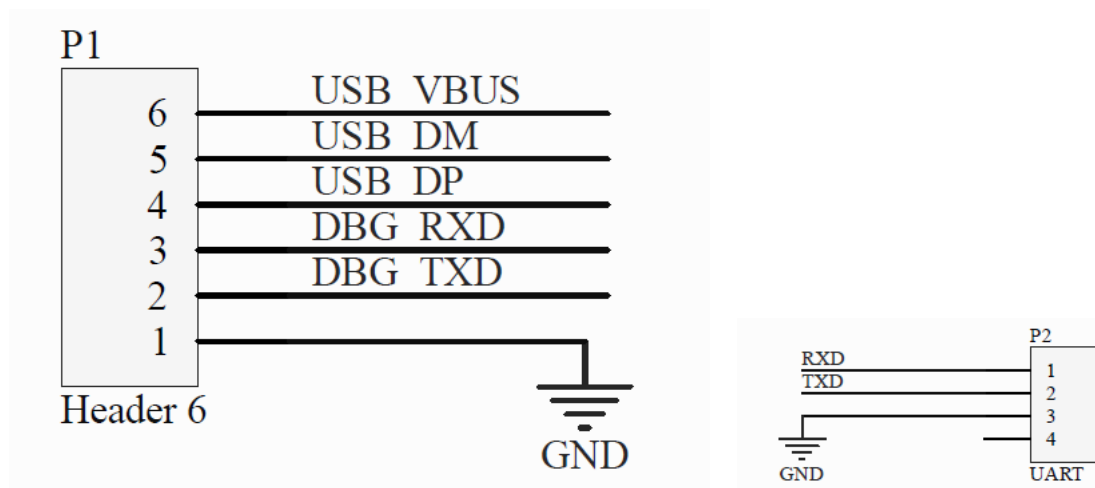


3.5. OTHERS

1. It is recommended to reserved a test point at RESET_N.

RESET_N	15	DI	Reset the module	$V_{ILmax} = 0.5\text{ V}$ $V_{nom} = 1.8\text{ V}$	Active low. A test point is recommended to be reserved if unused.
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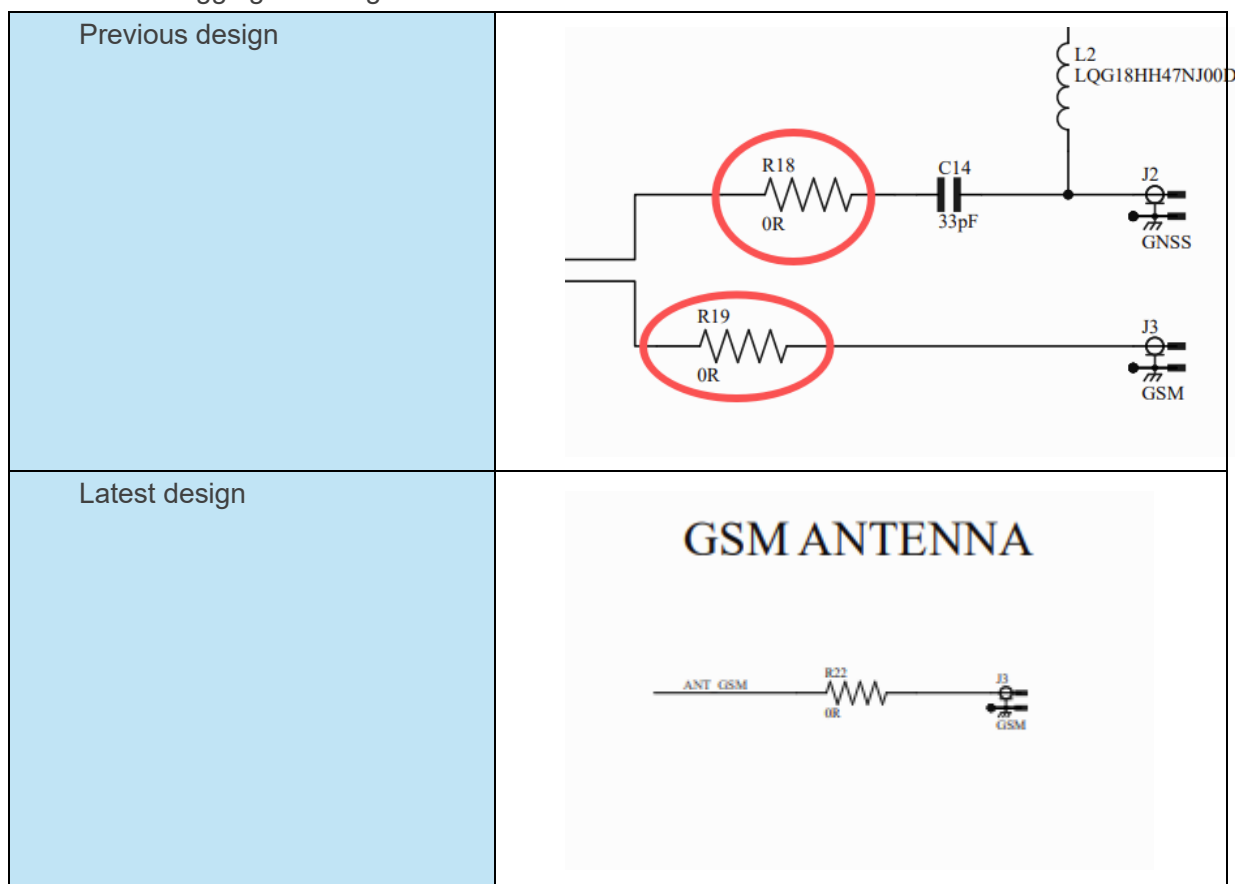
2. It is recommended to add a TVS diode near the connector for ESD protection.



3. It is recommended to add a TVS diode for ESD protection at all the test points.

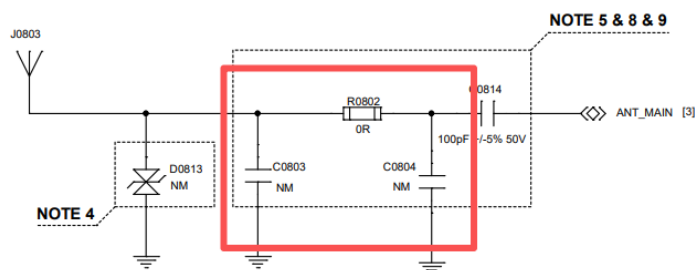
3.6. RF

- For 'GSM Antenna' trace, it is strongly recommended to reserve a Pi-matching circuit to facilitate debugging or tuning if needed.



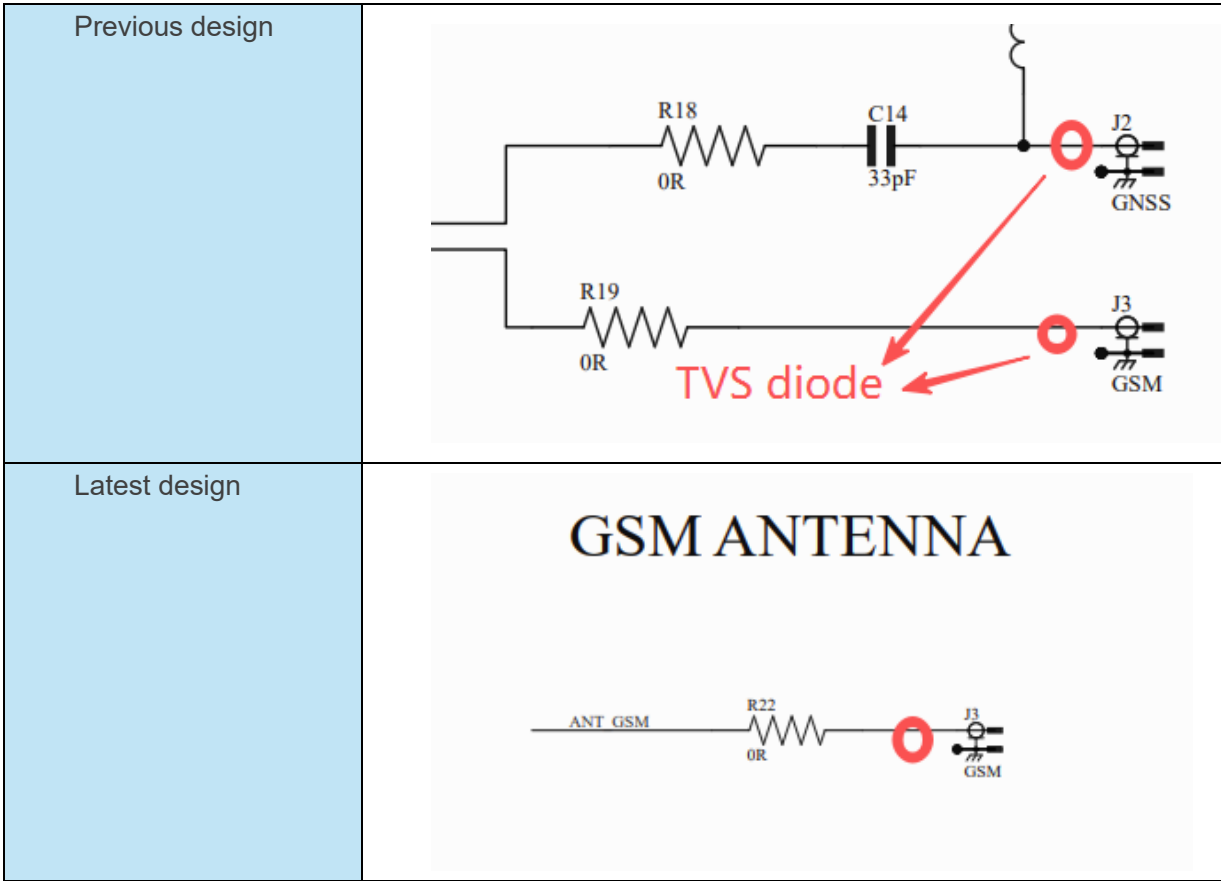
Reference design:

LTE/Wi-Fi Scan Antenna Design



*Please note that the 100 pF DC block capacitor (C0814) in the reference design is required when using DC power.

- For GSM antenna trace (ANT_MAIN), it is strongly recommended to reserve a TVS diode for ESD protection. Recommended to ensure that the capacitance junction of the TVS diode less than 0.05 pF for “ANT_MAIN” while “ANT_GNSS” antenna less than 0.5pF.



TVS diode used by customers based on schematic:

PESD Protection Devices

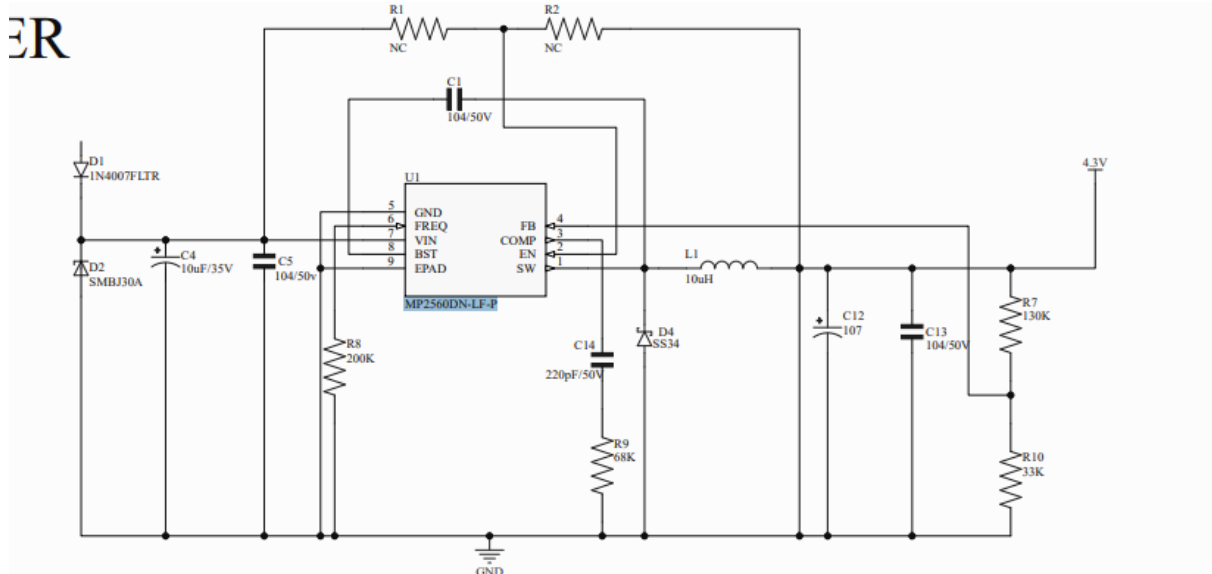
Table E1 — Electrical Characteristics

	Continuous Max Operating Voltage	Typ Trigger Voltage*	Typ Clamping Voltage†	Typ Capacitance @ 1 MHz, 1V _{RMS}	Typ Leakage Current	Max Leakage Current @ Max V _{DC}
Symbol	V _{DC}	V _{T(TYP)}	V _{C(TYP)}	C _P	I _{L(TYP)}	I _{L(MAX)}
Unit	V	V	V	pF	μA	μA
PESD0402-140	14	250	40	0.25	< 0.01	10.0
PESD0402-240	24	250	40	0.25	< 0.01	10.0
PESD0603-240	24	215	45	0.25	< 0.01	10.0
PESD1206Q-240	24	250	45	0.25	< 0.01	10.0

Notes : * TLP test method at 1kV.

- During layout design, please ensure the RF trace characteristic impedance would be 50Ω.

4. During layout design, it's strongly recommended to shield at DC-DC converter with RF shielding cover and reserve spacing for shielding frame.



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